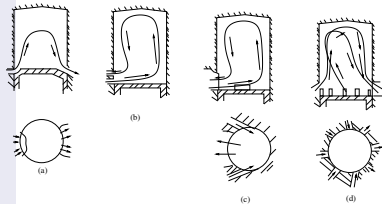


Scavenging in two stroke engines

Methods of Scavenging

- 1 Process of passing compressed charge through transfer port into engine cylinder flushing products of combustion
- 2 Crank case scavenging, separately scavenged engines -using blower or pump
- 3 Cross Scavenging -(a)
- 4 Loop Scavenging, MAN type-(b)
- 5 Loop Scavenging, Schnuerle type (c)
- 6 Loop Scavenging, Curtis type (d)

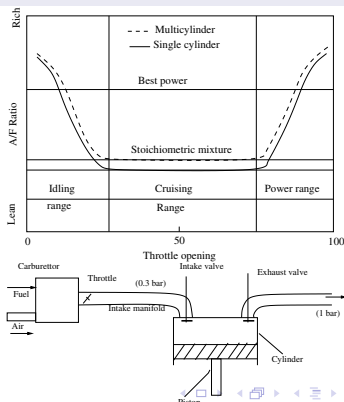


Fuel system

Carburetion

- 1 Process of formation of combustible fuel-air mixture by mixing proper amount of fuel with air
- 2 Require chemically correct mixture, lean mixture, rich mixture
- 3 Idling (Rich mixture), Cruising (lean mixture), High power (Rich mixture)

Carburettor performance and schematic of fuel system

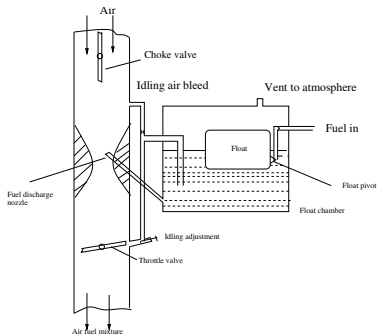


Carburettor

Simple Carburettor

- 1 Float and needle valve maintain level of gasoline
- 2 Pressure decreases at the venturi throat *carburettor depression*
- 3 gasoline engine quantity governed
- 4 simple carburettor fails to maintain A/F ratio

Simple Carburettor



Fuel system CI Engine

Fuel system

- 1 Fuel tank
- 2 Fuel filter
- 3 Fuel pump
- 4 Injector

Typical fuel system for a CI Engine

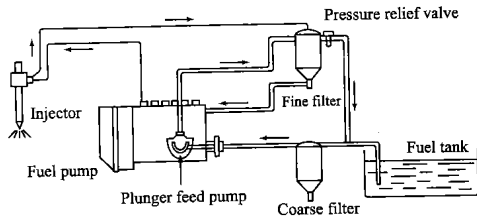


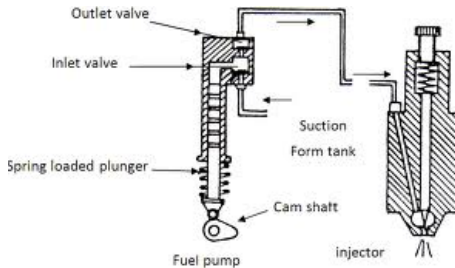
Fig. 9.1 Typical Fuel Feed System for a CI Engine

Fuel feed pump

Fuel feed pump

- 1 At minimum position, plunger creates suction
- 2 Cam turned to maximum lift position plunger goes up
- 3 Fuel delivered through outlet valve

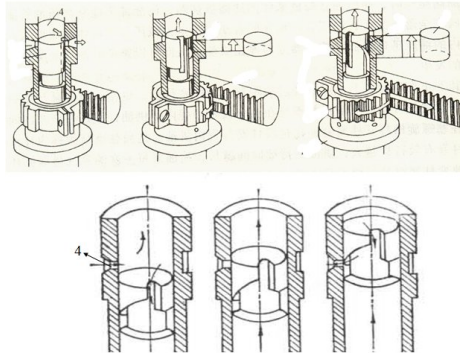
Schematic of fuel feed pump



Injection pump

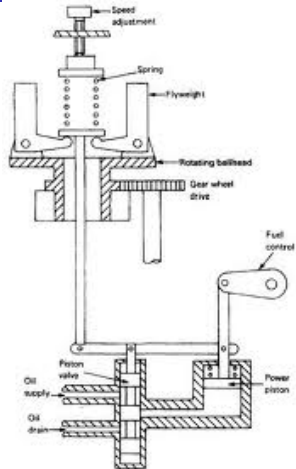
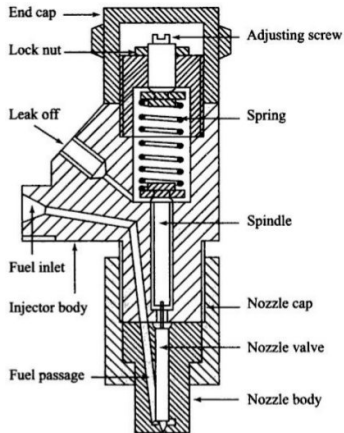
Fuel feed pump

- 1 Pump deliver metered quantity of fuel at high pressure, (120 to $200 \times 10^5 \text{ Pa}$)
- 2 Axial movement of plunger by Cam, rotation by rack
- 3 Fuel gets filled, as it rises and closes port -fuel injected



Injection pump

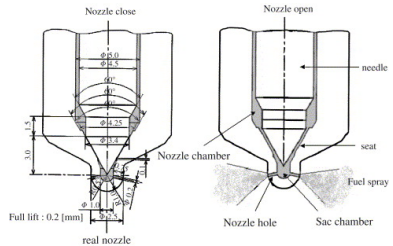
Bosch pump, Mechanical governor



Nozzles

Types of nozzles

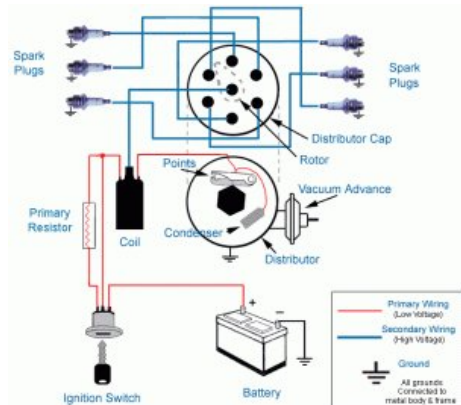
- 1 Pintle nozzles (8 -10 MPa)
- 2 Single hole nozzle (0.2 mm, 8 -10 MPa)
- 3 Multi hole (number 4 to 18, size 35 to 200 μ m, 18 MPa)
- 4 Pintaux Nozzle



Ignition

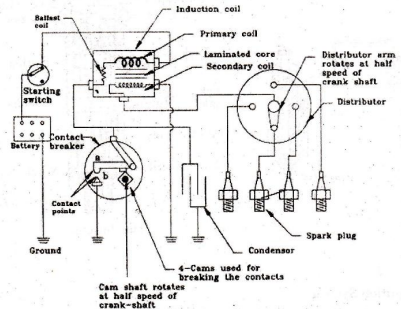
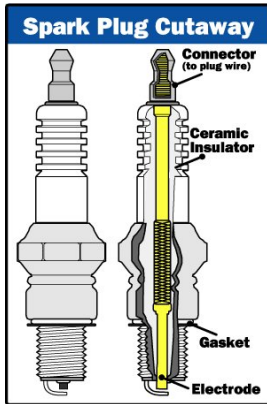
Battery Ignition

- 1 battery
- 2 ignition switch
- 3 ignition coil
- 4 contact breaker
- 5 capacitor
- 6 distributor
- 7 spark plug



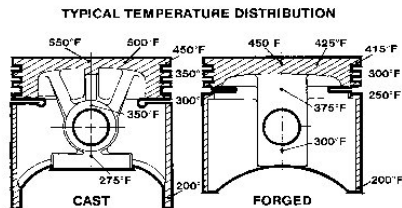
Spark plug

Ignition circuit schematic



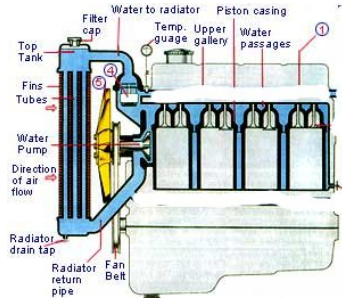
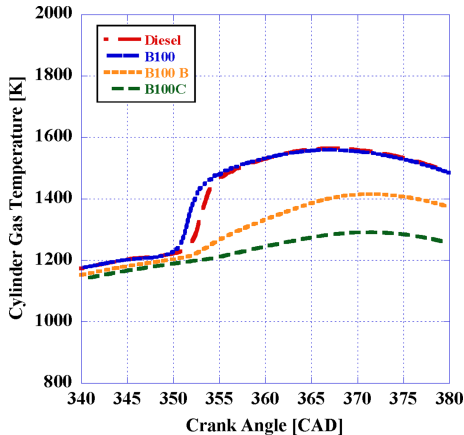
Engine cooling

Temperature variation



Engine cooling circuit

Temperature variation



Lubrication

- 1 Splash lubrication
- 2 Pressurised lubrication

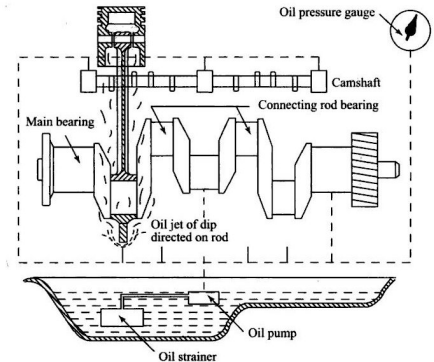
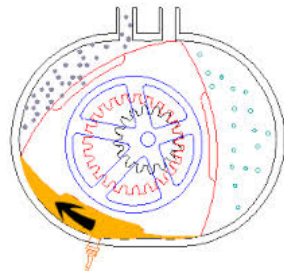


Fig. 13.10 Splash and Pressure Lubrication System

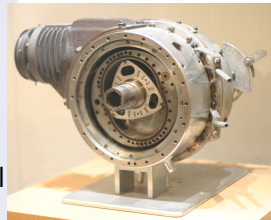
Wankel Engines

- 1 The Wankel engine is a type of internal combustion engine using an eccentric rotary design to convert pressure into rotating motion.
- 2 The engine was invented by German engineer Felix Wankel. He received his first patent for the engine in 1929.
- 3 Intake, compression, Ignition, Expansion, Exhaust.
- 4 Each combustion chamber in the Wankel engine generates one combustion stroke per driveshaft rotation. i.e, three power strokes per rotor rotation.



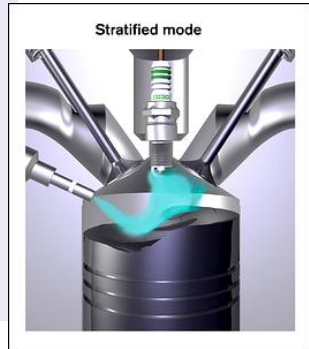
Wankel Engines

- 1 The Prime advantages of the Wankel engine are:
A far higher power to weight ratio than a piston engine. No reciprocating parts. Runs with almost no vibration. Not prone to engine-knock. Cheaper to mass-produce as contains few parts.
- 2 Disadvantages : Rotor sealing - problem - different temperatures in each separate chamber section. The combustion is slow as the combustion chamber is big and moving. Poor fuel consumption - exhaust stream is enriched with unburned mixture and carbon monoxide. Poor emissions.



Stratified charge engine

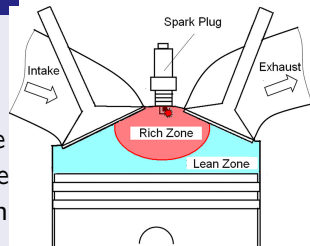
- 1 The fuel is injected into the cylinder just before ignition
- 2 This allows for higher compression ratios without "knock,"
- 3 Intake, compression, Ignition, Expansion, Exhaust.
- 4 Leaner air/fuel ratio than in conventional internal combustion engines.



Stratified charge engine

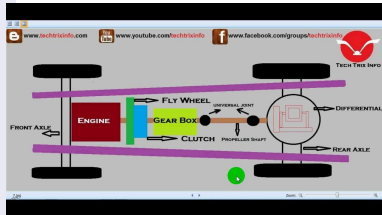
Disadvantages

- 1 Increased injector cost and complexity
- 2 Higher fuel pressure requirements
- 3 Carbon build-up on the back of the intake valve due to the lack of gasoline passing by the intake valve to act as a cleaning agent for the valve on traditional multi-port injection designs



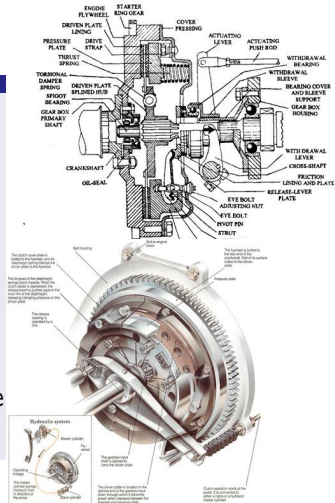
Transmission

- 1 A machine consists of a power source and a power transmission system
- 2 which provides controlled application of the power
- 3 Often transmission refers simply to the gearbox that uses gears and gear trains to provide speed and torque conversions from a rotating power source to another device



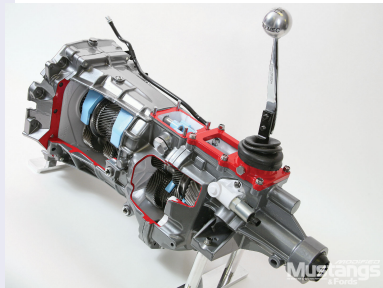
Clutch

- 1 A clutch is a mechanical device that engages and disengages the power transmission.
- 2 In the simplest application, clutches connect and disconnect two rotating shafts (drive shafts or line shafts)
- 3 Friction clutches, Centrifugal, Cone clutch



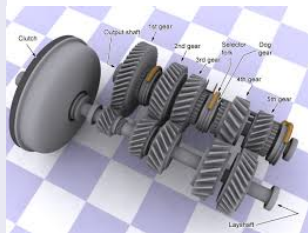
Gear Box

- 1 Most modern gearboxes are used to increase torque while reducing the speed of a prime mover output shaft (e.g. a motor crankshaft)
- 2 Engines typically operate over a range of 600 to about 7000 revolutions per minute
- 3 the car's wheels rotate between 0 rpm and around 1800 rpm.
- 4 Manual transmissions can include both synchronized and unsynchronized gearing



Gear Box

- 1 A simple but rugged sliding-mesh or unsynchronized/non-synchronous system, where straight-cut spur gear sets spin freely.
- 2 Synchronized by the operator matching engine revs to road speed, to avoid noisy and damaging clashing of the gears
- 3 Some commercial applications use non-synchronized manual transmissions that require a skilled operator
- 4 Automatic transmission that selects an appropriate gear ratio without any operator intervention



Differential

- 1 A differential couples the input shaft (or prop shaft) to the Pinion,
- 2 The differential gearing allows the outer drive wheel to rotate faster than the inner drive wheel during a turn.
- 3 A differential consists of one input, the drive shaft, and two outputs which are the two drive wheels

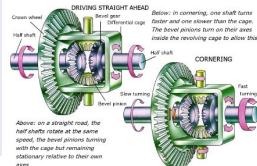
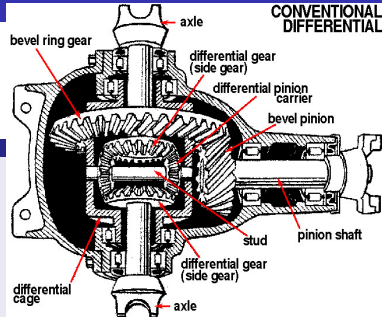
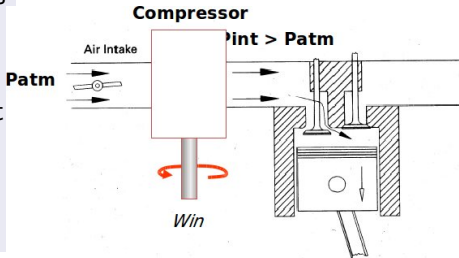


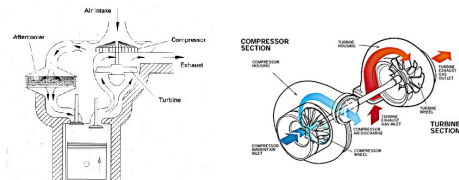
Fig 1: Differential - how it works

Supercharger

- 1 These devices are used to increase the power of an IC engine by raising the intake pressure and thus allowing more fuel to be burned per cycle.
- 2 Superchargers are compressors that are mechanically driven by the engine crankshaft and thus represents a parasitic load.



Turbocharger



- 1 Turbochargers couple a compressor with a turbine driven by the exhaust gas. The compressor pressure is proportional to the engine speed
- 2 Compressor also raises the gas temperature, so after-coolers are used after the compressor to drop the temperature and thus increase the air density.

Exhaust emissions

- 1 Oxides of nitrogen - Engine design, operating conditions, maximum at 14:1 A/F ratio, low at lean and rich mixture
- 2 Carbon monoxide - CO present in lean mixtures, high at idling, deceleration
- 3 unburnt hydrocarbons - incomplete combustion
- 4 Aldehydes Odourous oxygenated hydrocarbons - generally carcinogenic
- 5 Smoke

Emissions

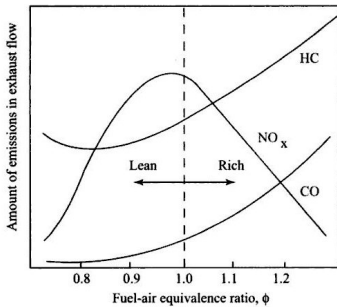


Fig. 15.1 Emissions as a Function of Equivalence Ratio for an SI Engine

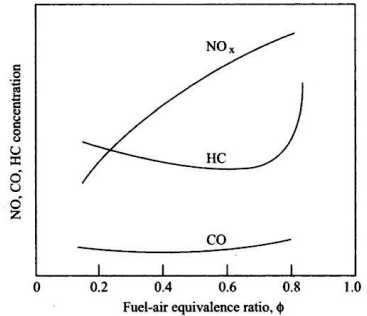


Fig. 15.2 Emissions as a Function of Equivalence Ratio for a CI Engine